

Multi-Source Operational Data Pipeline

The Operational Question

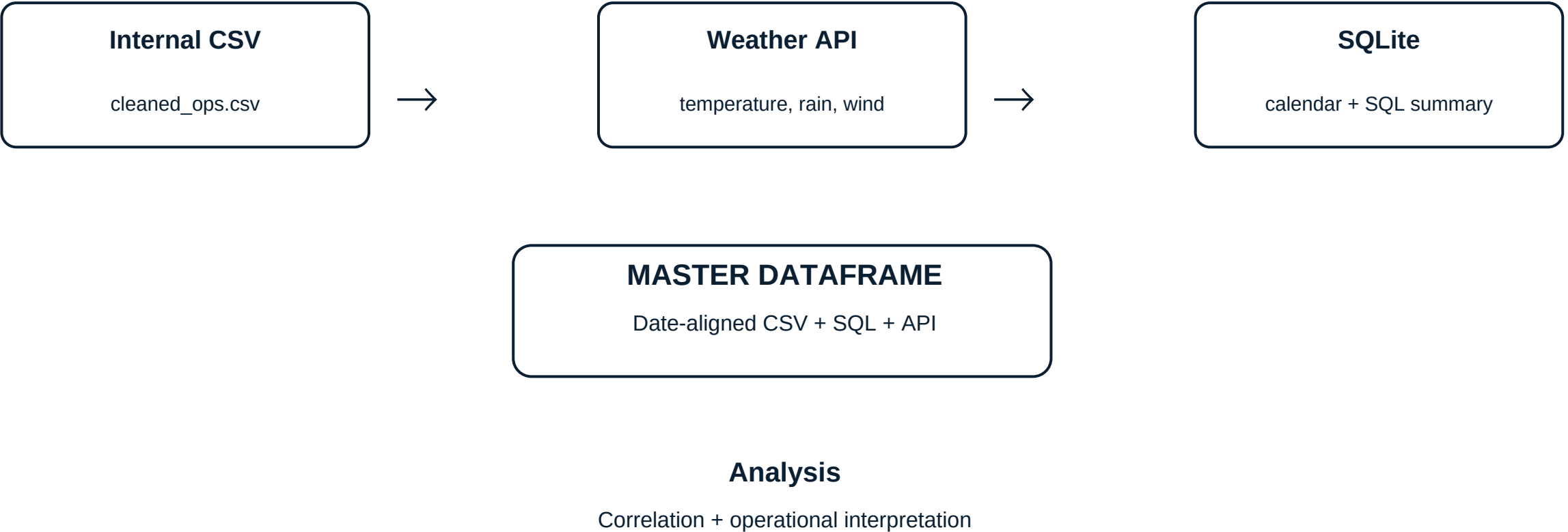
Can combining internal operational data with external weather data and a calendar database improve our understanding of operational performance?

Three sources

- Internal CSV: operational measurements
- External API: date-matched weather
- SQLite: calendar and operational summaries

Goal: create one reliable master dataset for operational analysis.

How the Data Was Integrated



SQL uses JOIN + GROUP BY and loads the result directly with `pd.read_sql()`.

Insight and Recommendation

Key finding

Pressure vs Flow: $r \approx -0.009$

The relationship is essentially linear-neutral in this dataset. Pressure alone does not explain changes in flow rate.

Recommendation

- Do not rely on pressure alone when explaining flow-rate changes.
- Add equipment status, maintenance events, staffing and production schedules.
- Continue collecting date-aligned external conditions for future analysis.

Note: correlation does not establish causation, and the operational date range is limited.